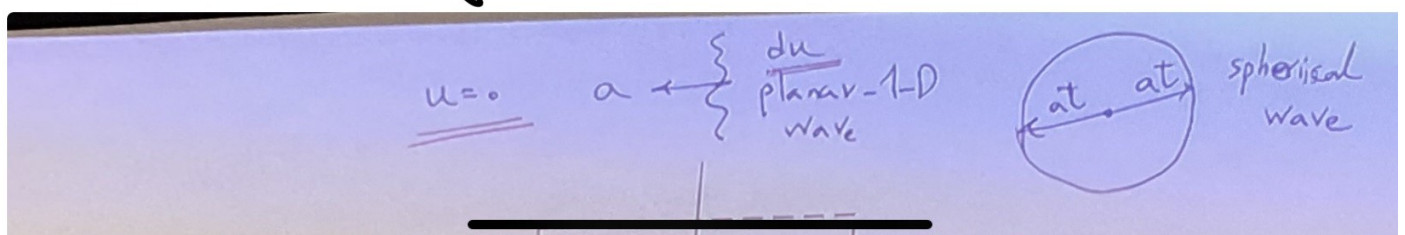
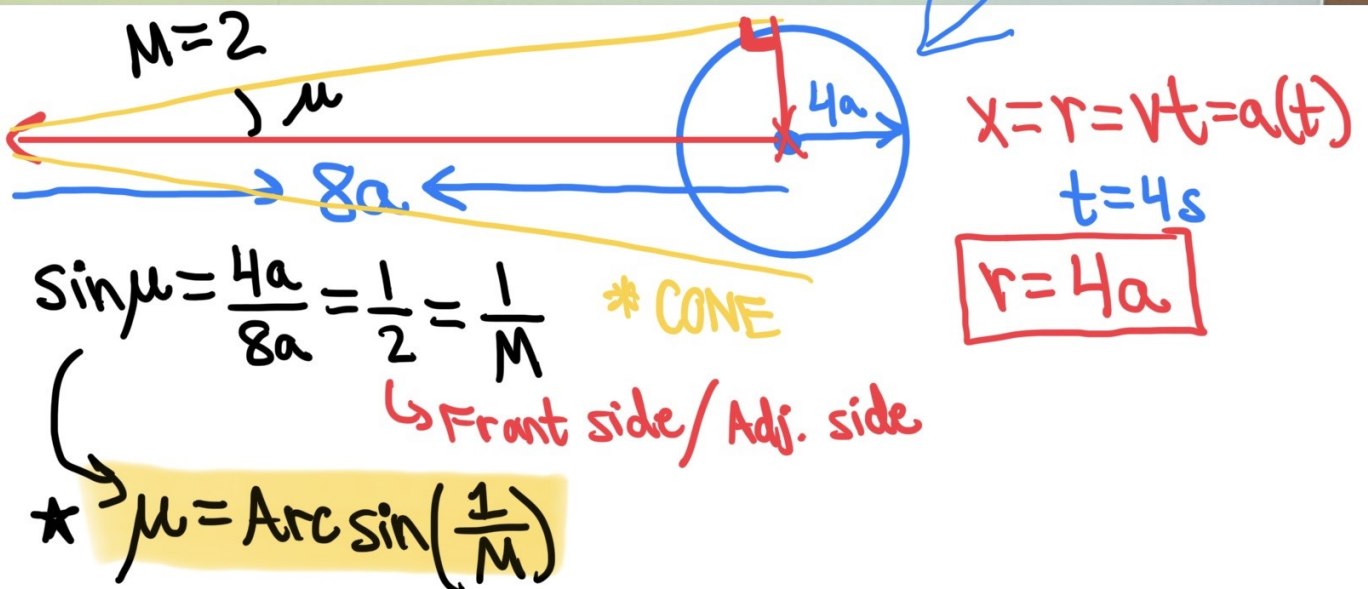
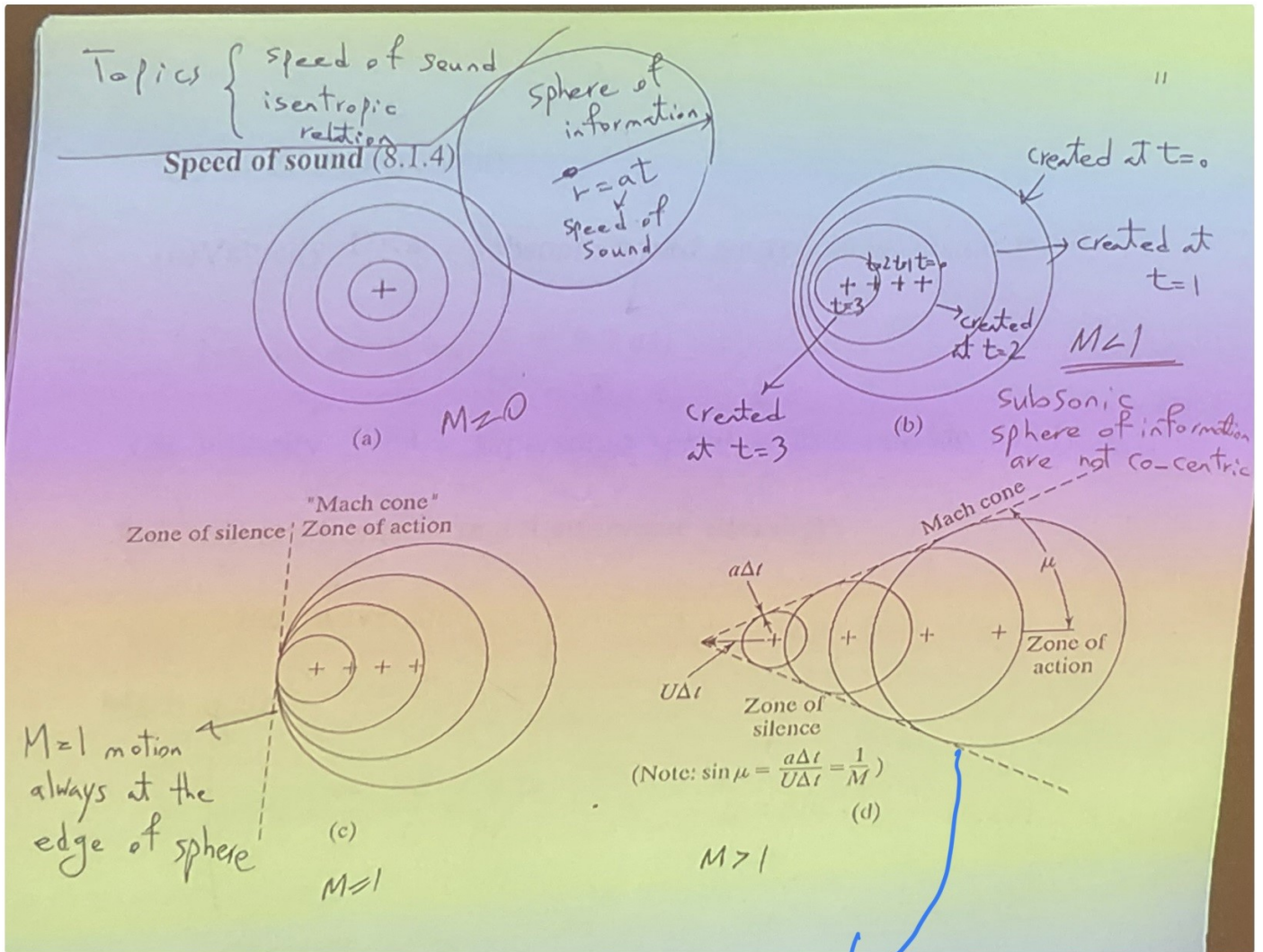
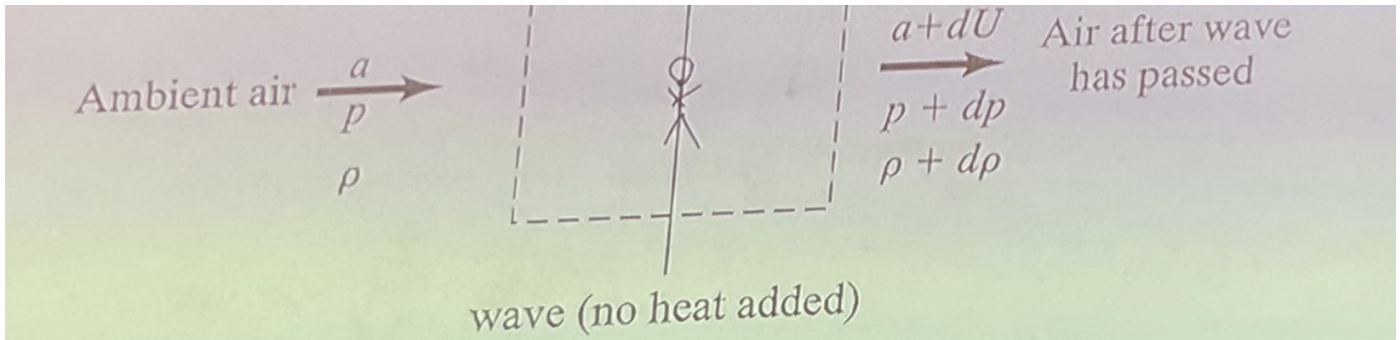




(SEE Ad

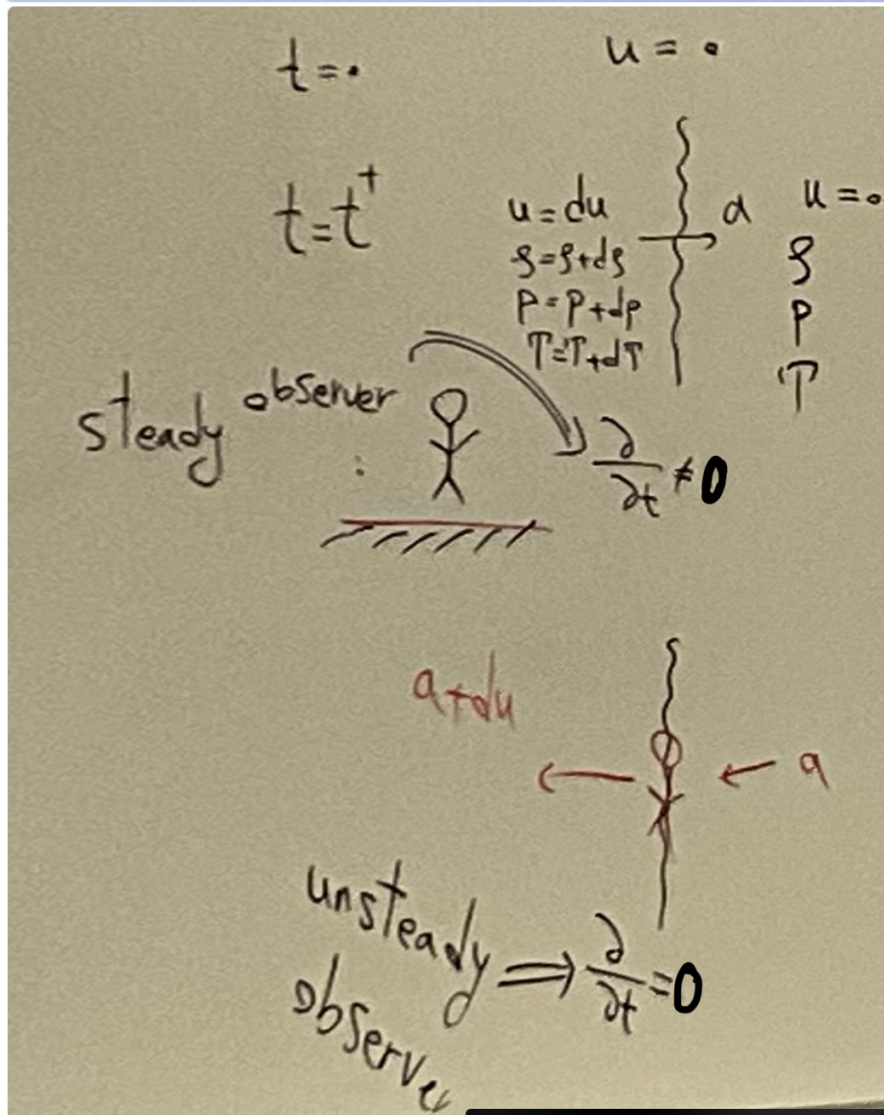




assume a weak wave with speed a ; changes to speed, pressure density, temperature are small $\Rightarrow$ isentropic

continuity: $\rho a = (\rho + d\rho)(a + dU)$ (8.23)

momentum: $p + \rho a^2 = (p + dp) + (\rho + d\rho)(a + dU)^2$ (8.24)

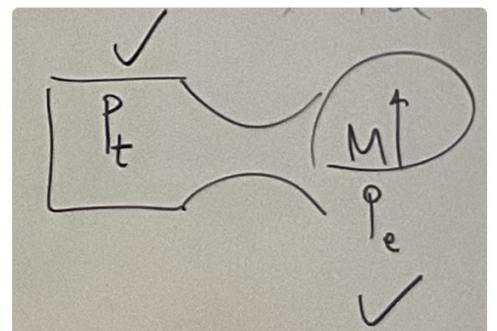


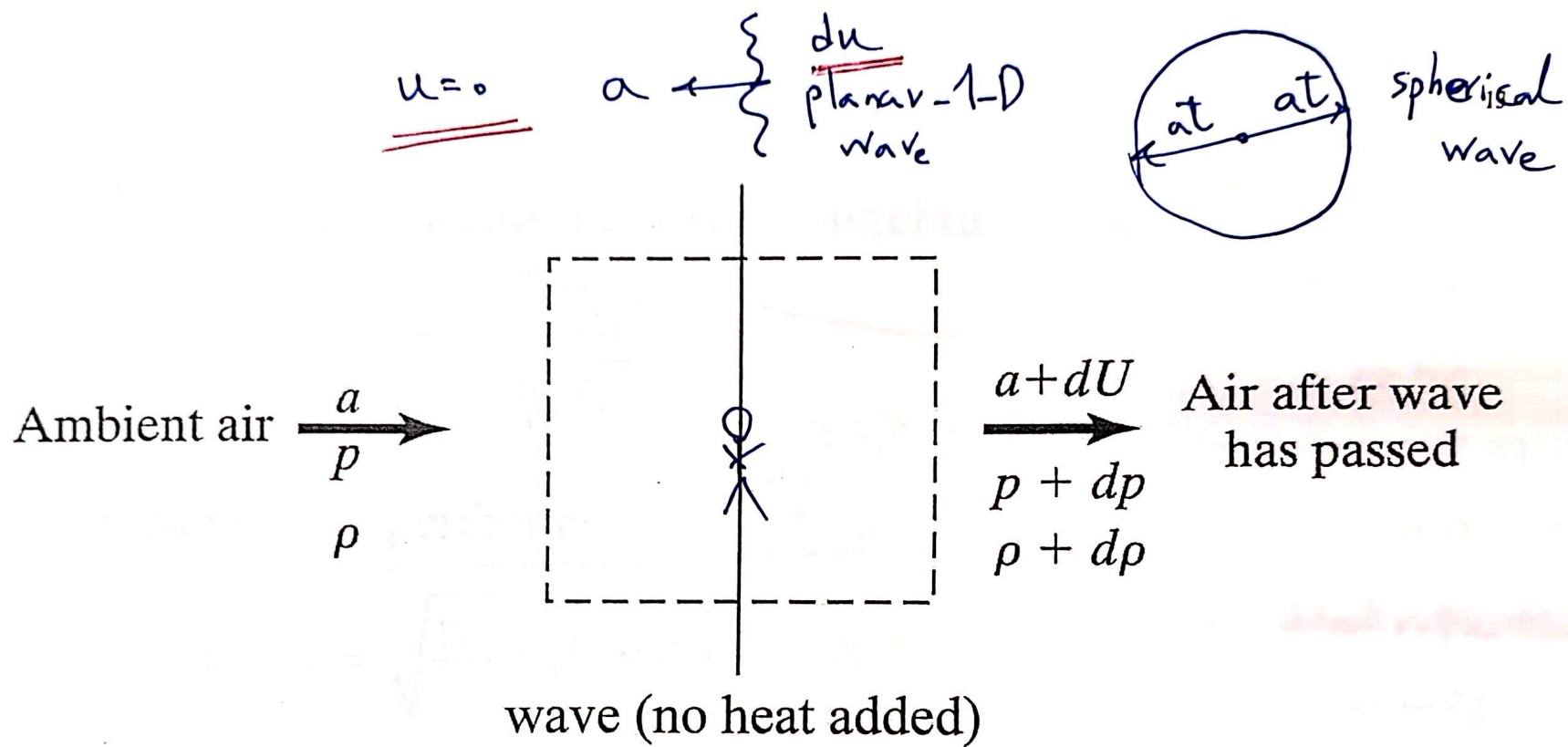
* Incomp.

$$P + \frac{1}{2} \rho u^2 = \text{const.}$$

* Comp.

$$P + \rho u^2 = \text{const}$$





assume a weak wave with speed a ; changes to speed, pressure density, temperature are small $\Rightarrow$ isentropic

continuity: $\underline{\rho a} = (\rho + d\rho)(a + dU)$ (8.23)

momentum: $p + \rho a^2 = (p + dp) + (\rho + d\rho)(a + dU)^2$ (8.24)

ignoring 2nd order terms, after some algebra

$$\Rightarrow a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} \text{ - isentropic}$$

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_{s=\text{cte}}} \quad (8.26)$$

for a calorically perfect gas:

$$p = \rho^\gamma \times \text{constant}$$

$$\Rightarrow a = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma R T}$$

$$p = \text{cte } \rho^\gamma$$

Isentropic flow
(8.28)

for standard sea level: $a_s = 340.9 \text{ m/s} = 1117 \text{ ft/s}$

$$M \equiv \frac{U}{a}$$

↓

Mach number

$$M < 1 \text{ subsonic}$$

$$M = 1 \text{ sonic}$$

$$M > 1 \text{ supersonic}$$

for incompressible flow:

$$M = 0 \quad ; \quad a \rightarrow \infty$$

Assume Adiabatic $\rightarrow q = 0$ No viscous heat work

Energy equation: First law $\dot{m}(du_e + w) = 0 \rightarrow (u_e + w)_{\text{Inlet}}^{\dot{m}} = (u_e + w)_{\text{outlet}}^{\dot{m}}$

$P_1 A_1 = F_1$
 $F_1 U_1 = \text{Power}$
 $\underbrace{P_1 U_1 A_1}_{\text{rate of work} \rightarrow \text{Power}} + \underbrace{u_{e1} \rho_1 U_1 A_1}_{\text{internal Eng.}} + \underbrace{\left(\frac{U_1^2}{2}\right) \rho_1 U_1 A_1}_{\text{k.E.}} = \underbrace{P_2 U_2 A_2}_{\text{Power}} + \underbrace{u_{e2} \rho_2 U_2 A_2}_{\dot{m} u_e} + \underbrace{\frac{U_2^2}{2} \rho_2 U_2 A_2}_{\dot{m} \text{ k.E.}} \quad (*)$

Using continuity

$$\rho_1 U_1 A_1 = \rho_2 U_2 A_2 \quad (1)$$

and enthalpy definition

$$h = u_e + \underbrace{p \nu}_{\frac{p}{\rho}} \quad (\nu = 1/\rho) \quad (2)$$

$$\Rightarrow \boxed{H_t = h_1 + \frac{U_1^2}{2} = h_2 + \frac{U_2^2}{2}} \quad (8.29)$$

$H_t \equiv$ total (stagnation) enthalpy

For a perfect gas:

$$c_p T_1 + \frac{U_1^2}{2} = c_p T_2 + \frac{U_2^2}{2} \quad (8.30)$$

$$T_{t1} = T_1 + \frac{U_1^2}{2c_p} = T_2 + \frac{U_2^2}{2c_p} = T_{t2} \quad (8.31)$$

$$T_t = T + \frac{U^2}{2c_p} = T_{static} + T_{dynamic} = \text{constant} \quad \text{if } q = 0, \mu = 0 \quad (8.32)$$

$$T_{t1} = T_{t2} \quad (8.33)$$

(adiabatic process)

$$T_t = T + \frac{U^2}{2c_p} = T + \frac{M^2 a^2}{2} = T + \frac{M^2 \cancel{\gamma} R T}{2 \cancel{\gamma} R} = T + \frac{\gamma - 1}{2} M^2 T \Rightarrow$$

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2$$

for adiabatic flow

$$\frac{T_t}{T} = \left(\frac{P_t}{P} \right)^{\frac{\gamma-1}{\gamma}}$$

From (8.32) using equation 8.16 for c_p

$$\frac{T_t}{T} = 1 + \frac{U^2}{2\gamma RT / (\gamma - 1)} = 1 + \frac{\gamma - 1}{2} M^2$$

$$\frac{P_t}{P} = \left(\frac{T_t}{T} \right)^{\frac{\gamma}{\gamma-1}} \quad (8.34)$$

[Can be also derived starting momentum equation for isentropic inviscid flow-see text]

For isentropic flow:
(using 8.20)

$$\frac{P_t}{P} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\gamma/(\gamma-1)} \quad (8.36)$$

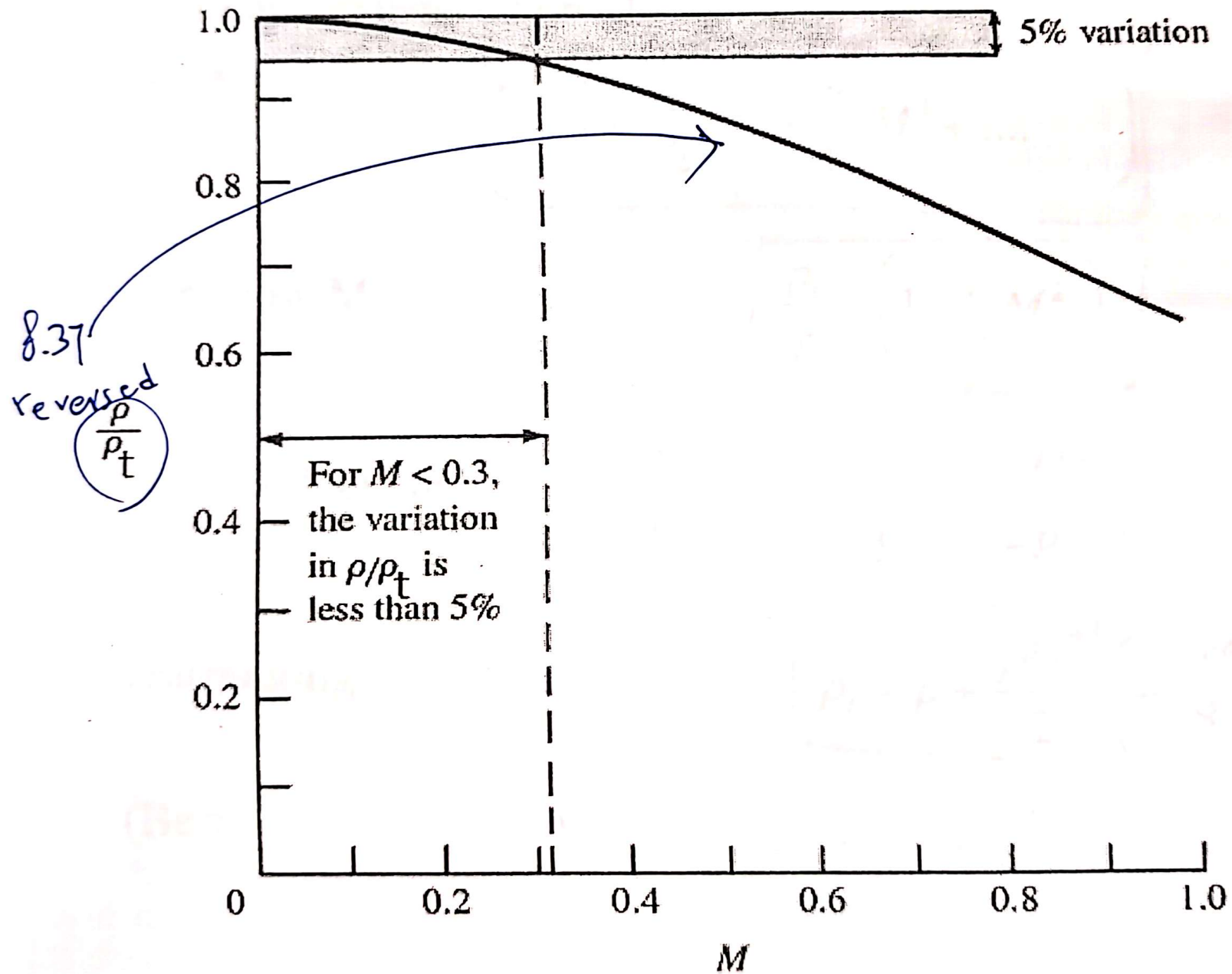
Similarly for density

$$\frac{\rho_t}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{1/(\gamma-1)} \quad (8.37)$$

p_t and ρ_t are stagnation (total) pressure and density

from (8.36) 5% error in density for $M = 0.32$ 10% for $M = 0.44$
1% for $M = 0.14$

8.34, 8.36, 8.37 are tabulated in Table 8.1 (isentropic tables)



$$(1+x)^n = 1 + nx + \frac{n}{2}x^2 + \dots \quad |x| \ll 1$$

$$M^2 \ll 1$$

binomial expansion on M^2 :

(for small M^2)

$$\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{8} M^4 + \dots \right) \quad (8.38)$$

for a low M :

$$\frac{p_t}{p} = \left(1 + \frac{\gamma}{2} M^2 \right) \quad (8.39)$$

for a perfect gas:

$$\frac{p_t}{p} = 1 + \frac{\cancel{M^2} U^2}{2 \cancel{RT}} = 1 + \frac{U^2}{2p/\rho} \quad (8.40)$$

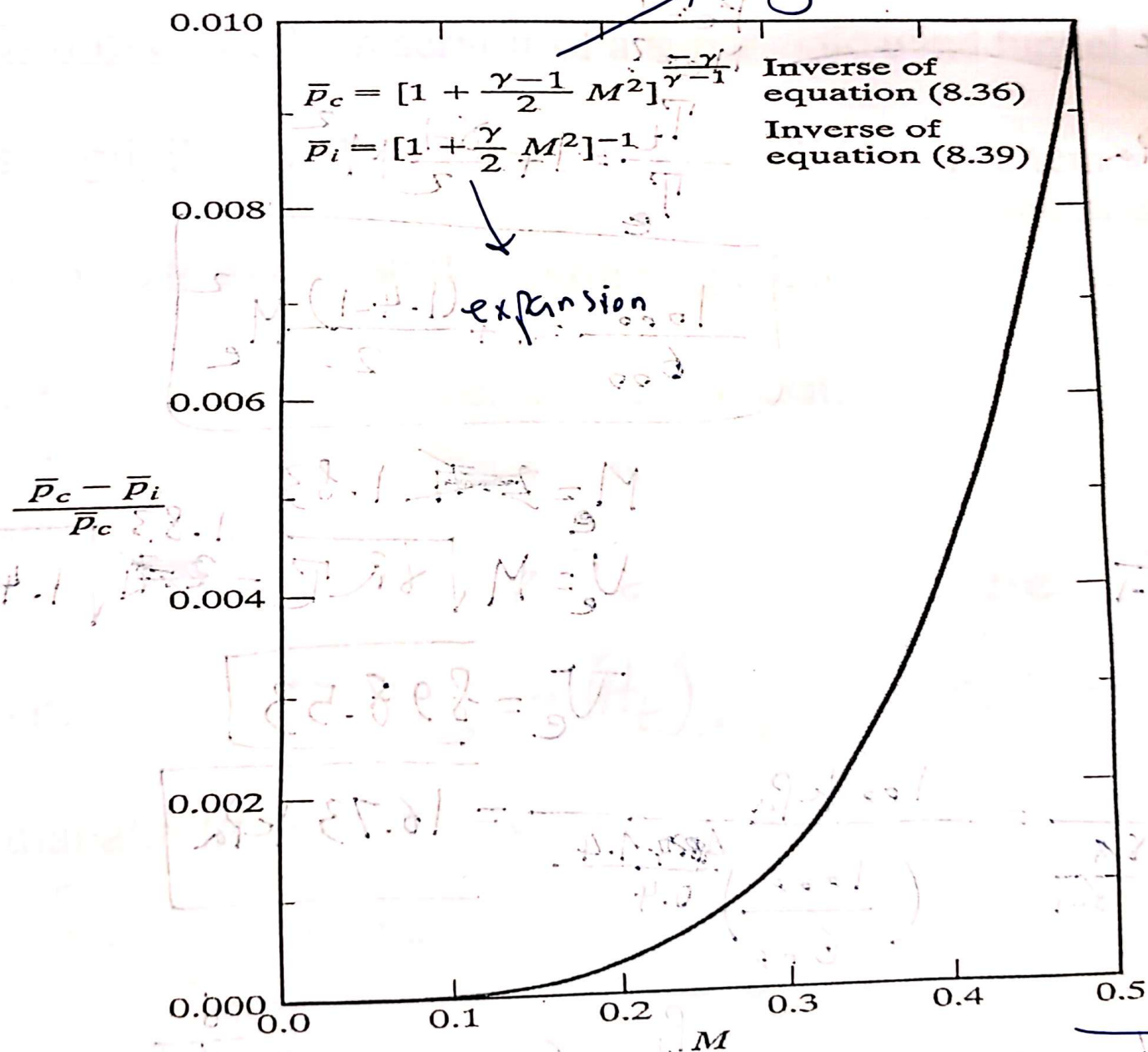
rearranging:

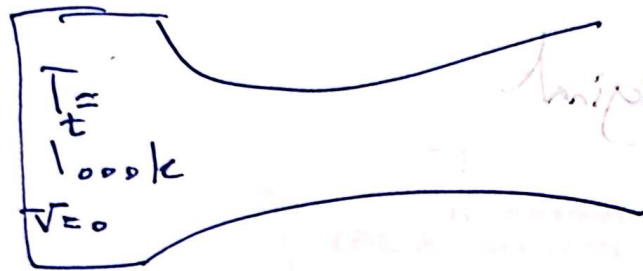
$$p_t = p + \frac{\rho U^2}{2} \left(1 + \frac{\gamma}{8} M^4 + \dots \right) \quad (8.41)$$

(Bernoulli's equation)

$$\text{error} \approx 1\% \text{ for } M = 0.5 \quad \underline{\underline{0.5}}$$

$$\underline{\underline{0.15\% \text{ for } M = 0.3}}$$





assume: $P = 100 \text{ kPa}$

$$T_e = 600 \text{ K}$$

$$P_e = ?$$

$$M_e = ?$$

$$\frac{T_t}{T_e} = 1 + \frac{\gamma - 1}{2} M^2$$

$$\boxed{\frac{1000}{600} = 1 + \frac{(1.4 - 1)}{2} M_e^2}$$

$$M_e = \cancel{2.71} 1.83$$

$$V_e = M \sqrt{\gamma R T_e} = \cancel{2.71}^{1.83} \sqrt{1.4 \times 287 \times 600}$$

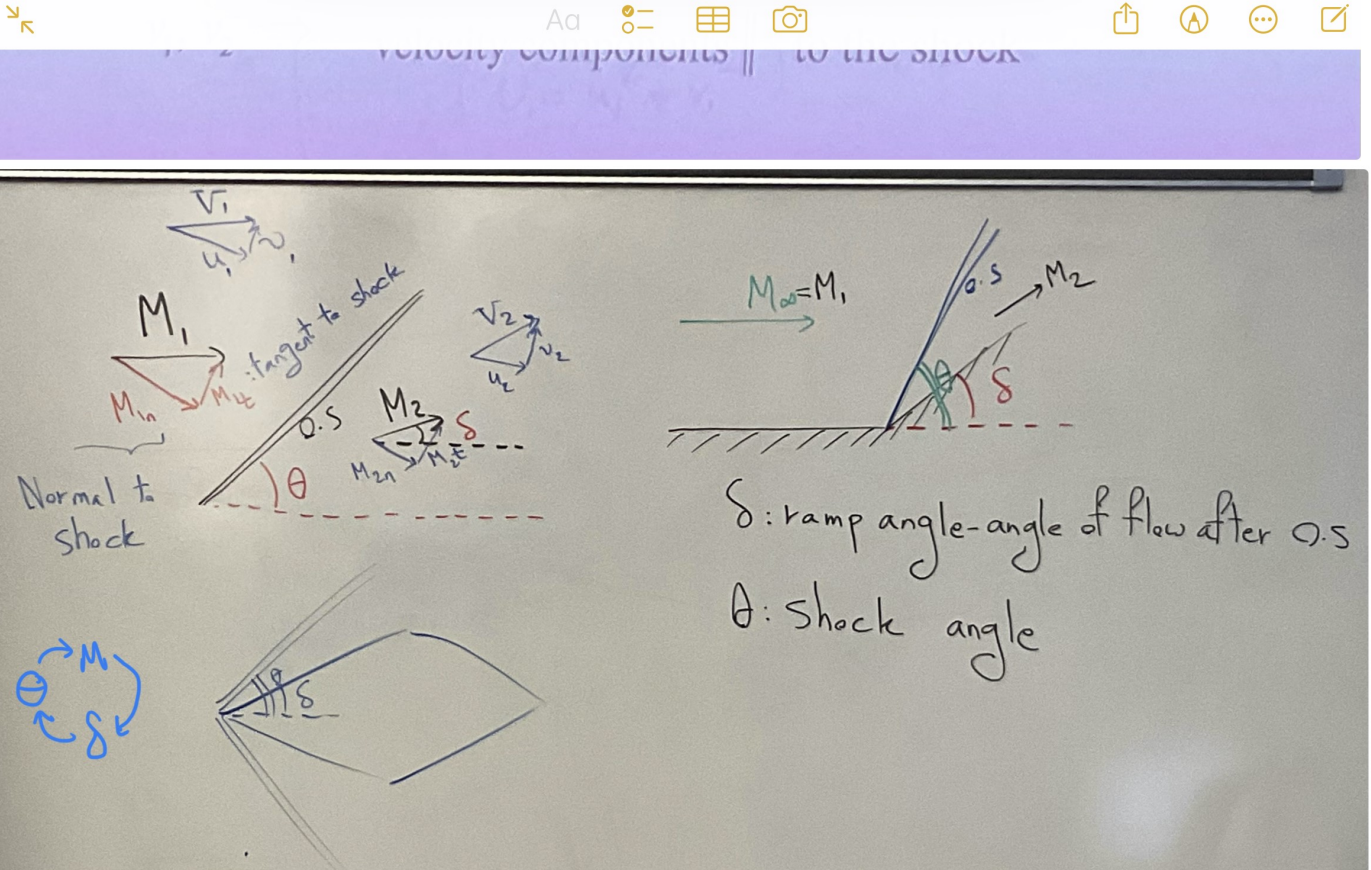
$$\boxed{V_e = 898.53}$$

$$P_e = \frac{P_t}{\left(\frac{T_t}{T_e}\right)^{\frac{\gamma}{\gamma-1}}} = \frac{100 \text{ kPa}}{\left(\frac{1000}{600}\right)^{\frac{1.4}{0.4}}} = \boxed{16.73 \text{ kPa}}$$

You can use

this as well:

$$\frac{P_t}{P_e} = \left(1 + \frac{\gamma - 1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$$





Aa



★ (SEE NOTES POSTED)

